Report - Version 14: Instruction Tuning

Introduction

Version 14 preserves the complete Version 13 experiment and adds a second learning stage: supervised instruction tuning. The WikiText/BPE base model is still selected by validation loss, then a small deterministic subset of Databricks Dolly 15K is used to adapt the same 34,496 parameters.

The new educational mechanism is response-focused masked loss. Prompt tokens remain visible to causal attention, but only response targets contribute to the supervised objective. Dolly validation selects a separate BEST instruction checkpoint before held-out instruction evaluation.

Version 14 base model restored from BEST

```
-> Instruction / optional Context / Response formatting
-> response-focused sequence windows
-> masked supervised fine-tuning
-> BEST instruction checkpoint
-> held-out instruction test
-> base vs tuned controlled generation
```

Goal of Version 14

- preserve the Version 13 base model, held-out evaluation, and controlled generation
- use a deterministic Dolly subset split 192 / 32 / 32
- make prompt and response formatting explicit
- restrict supervised loss to response targets
- select the BEST instruction checkpoint using masked validation loss
- compare base and tuned models with the same decoding procedure

Architecture and continuity

```
preserved WikiText/BPE base pipeline
-> BEST base model
-> Databricks Dolly 15K
-> selected 256 rows: 192 train / 32 validation / 32 test
-> explicit prompt formatting
-> response mask on 16-token BPE windows
-> 8 SFT epochs, learning rate 0.15
-> BEST instruction -> held-out test
```

Continuity with Version 13

WikiText subsets, BPE vocabulary 2,048, the two-block 34,496-parameter Transformer, LAST/BEST base checkpoints, 20 base epochs, held-out base evaluation, and controlled generation remain unchanged. The tokenizer is reused and is not retrained on Dolly.

Instruction dataset

Setting	Value
Dataset	Databricks Dolly 15K
Selected rows	256
Train / validation / test	192 / 32 / 32
Batch size	32
Learning rate	0.15
Fine-tuning epochs	8
Max useful windows / example	2
Generation tokens	16

Missing Dolly input stops explicitly with FileNotFoundError; the notebook does not download or silently substitute another instruction corpus.

Formatting and supervised targets

Each Dolly record is converted into one explicit text sequence. Context is included only when non-empty. The prompt tokenization must be an exact prefix of the complete example.

Field	Template
Instruction	Instruction: {instruction}
Context	Context: {context} (optional)
Response	Response: {response}

Response mask

Prompt tokens remain part of the causal context, but prompt target positions receive mask 0 and response target positions receive mask 1. The training loss is token-weighted only over response targets.

```
masked_loss = sum(token_loss * response_mask)
/ sum(response_mask)
```

Window construction

- fixed 16-token BPE context
- first useful window starts far enough back to preserve prompt conditioning before the response
- windows with no response targets are discarded
- at most two useful windows are retained per instruction example

Masked supervised fine-tuning

instruction_model is created as a fresh model and initialized by copying the parameters of the BEST base model exactly. Fine-tuning updates the same 34,496 trainable parameters with GradientTape and a manual learning rate of 0.15.

Setting	Value
Instruction batch size	32
Instruction learning rate	0.15
Instruction epochs	8
Shuffle seed	SEED + 1000 + epoch
Printed epochs	1, 4, 8
Selection criterion	lowest masked validation loss

Instruction checkpoint

Only BEST is retained for the instruction stage: model + best_epoch + best_validation_loss. After 8 epochs, a fresh instruction_tuned_model restores the checkpoint with the lowest masked validation loss.

Final instruction evaluation

Masked train and validation losses are recomputed from the restored tuned model. Dolly test loss is measured only after selection, and instruction-test perplexity is reported as exp(masked test loss).

The inherited Version 13-style base evaluation remains separate egins.

Base vs instruction-tuned generation

One example is selected from the Dolly test split and converted to the same Instruction / optional Context / Response prompt for both models. The reference response is printed only for inspection; it is not used during generation.

Model	Decoding	New tokens
base BEST	greedy + top-p 0.9	16
instruction-tuned BEST	greedy + top-p 0.9	16

The comparison demonstrates the mechanics of adaptation, not a guaranteed semantic improvement. The notebook does not compute instruction-adherence, helpfulness, safety, or human-preference scores.

Included checks

- instruction train/validation/test source rows are disjoint
- prompt formatting starts with Instruction and ends with Response
- masks are binary and every retained window contains response targets
- the instruction model starts exactly from base BEST
- fine-tuning changes at least one parameter
- a real BEST instruction checkpoint is selected and restored
- all masked losses and perplexity values are finite
- restored validation loss matches BEST metadata
- base/tuned generation lengths are correct
- seeded top-p tuned generation is reproducible
- causal attention remains normalized and future-masked
- the final test cell reports All checks passed

What Version 14 teaches

Version 14 shows that instruction tuning does not require a new architecture. The same decoder-only Transformer can be adapted by changing the supervised data, making prompt structure explicit, and restricting the objective to response targets.

Version 14 base BEST

```
-> transparent prompt
-> response mask
-> supervised SFT
-> BEST instruction
-> held-out test + base/tuned comparison
```

Limitations

- only 256 Dolly records are selected
- the inherited 16-token context truncates long instruction prompts
- the BPE tokenizer is not retrained on Dolly
- the objective is response-only supervised loss with no preference optimization
- RLHF, DPO, PPO, tool use, and agents are not included
- the goal is to expose SFT mechanics, not to produce a practical assistant

Conclusion

Version 14 adds a compact supervised instruction-tuning stage without changing the base architecture. Prompt formatting, response masking, validation-selected instruction checkpointing, held-out instruction evaluation, and base-versus-tuned generation make the complete adaptation path visible.

Future versions will introduce new components and gradually evolve the architecture.